Executive Brief: Reducing Customer Churn

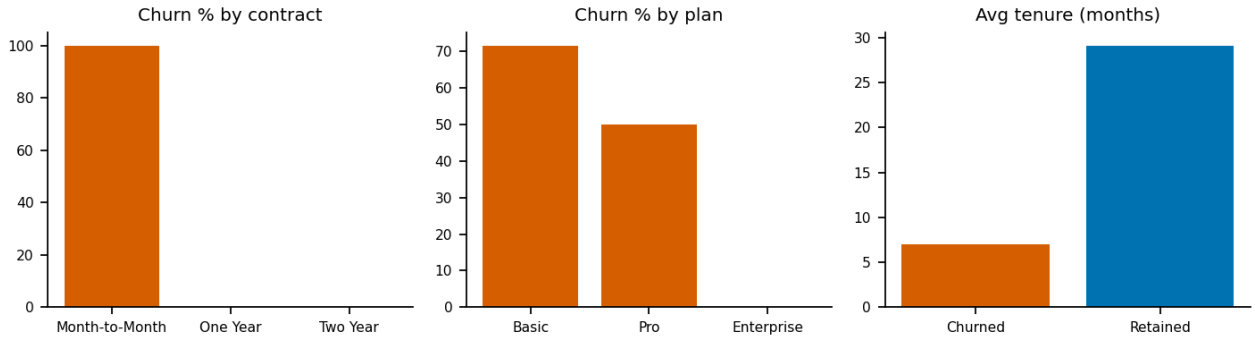
Data Analytics Capstone | Dataset: 15 synthetic customers

The problem

7 of 15 customers (46.7%) have churned, putting **$410 of $1,270 monthly recurring revenue (32%)** at risk. The business needs to know who leaves, why, and what to do first.

The answer in one line

Churn is concentrated in **new, Month-to-Month, Basic/Pro customers who raise 3 or more support tickets**. Every customer on a One Year or Two Year contract stayed.



Narrative: problem (46.7% churn) → where it concentrates (contract, tenure, tickets) → what it costs (up to $410/mo) → what to do (early engagement and annual-plan conversion) → how sure we are (small sample, treat as indicative).

Five insights that affect the decision

#	Insight	Evidence	Decision it supports
1	Contract type is the strongest churn signal	Month-to-Month: 7/7 churned. One Year and Two Year: 0/8 churned.	Prioritise moving customers to longer contracts.
2	Churn happens early	Churned customers averaged 7.0 months tenure vs 29.1 for retained. All churned left within 14 months.	Focus retention effort on the first 12 months.
3	Support friction predicts churn	Customers with 3+ tickets: 7/7 churned (avg 4.3 tickets). Under 3 tickets: 0/8 churned.	Trigger outreach when a customer hits 3 tickets.
4	Lower-value plans leave most	Basic 71% churn, Pro 50%, Enterprise 0%. Enterprise brings $600 of $1,270 MRR.	Protect Enterprise; upgrade-path offers for Basic/Pro.
5	Risk score separates customers well	Rule-based score (0-100): all 7 High-risk churned; no Low-risk churned.	Use the score to prioritise outreach lists.

Customer segments

Segment	Customers	Churn	Monthly revenue
New & Flexible (Month-to-Month, tenure up to 12m)	6	100%	$330
Established Flexible (Month-to-Month, over 12m)	1	100%	$80
Committed Standard (One/Two Year, Basic/Pro)	4	0%	$260
Loyal Enterprise (Two Year)	4	0%	$600

Expected impact (ranges and assumptions)

Scenario	Share of at-risk revenue saved	Monthly revenue protected	Annual
Low	20%	$82	$984
Mid	35%	$143	$1,722
High	50%	$205	$2,459

Assumptions

• At-risk revenue = $409.93/month from the 7 churned customers (all Month-to-Month). The same pattern is assumed to repeat for a similar group of new customers.

• A retention programme (onboarding check-in, ticket outreach, annual-plan offer) saves 20-50% of those customers. This is a planning assumption, not a measured result.

• Revenue means monthly charges only. Costs of the programme are not included.

Recommendations

• **New & Flexible:** 30-day onboarding check-in and an annual-plan incentive.

• **Support friction:** proactive outreach and priority resolution once a customer reaches 3 tickets.

• **Basic/Pro:** upgrade-path offer after the first 6 months.

• **Loyal Enterprise:** protect with account reviews and early renewal offers (40%+ of revenue).

Limitations and next steps

• Only 15 synthetic customers. The perfect separation (100% vs 0%) is almost certainly overfit and will not hold on real data.

• Contract, tenure, plan and tickets overlap, so effects cannot be separated. Correlation is not proof of cause.

• Thresholds (for example 3 tickets, 12 months) are judgment-based.

• Next steps: test on a larger dataset, run a small pilot with a control group, and recalibrate the score.